

Impact of Weather Severity on Taxi Drivers' Earnings Potential Across New York City

MAST30034 Project 1

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Github repo with commit

August 31, 2026

1 Introduction

In the large, busy and competitive taxi landscape of New York City (NYC), it is critical for taxi drivers to make informed decisions to maximise their earning opportunities. A key factor that can make these decisions more challenging is the weather and its severity. Its unpredictable nature can influence travel conditions and passenger demand which may introduce nuances in how drivers adjust their strategies. This report investigates how weather severity relates to taxi drivers' earnings potential across different locations and times. To examine this relationship, we consider differences across pickup location, time of day, day of week and precipitation type. We aim to provide NYC Yellow Taxi drivers with evidence-based recommendations for planning their work during adverse weather. This driver population is chosen due to the broader spatial coverage and greater prevalence of Yellow Taxis in NYC. Green Taxis were excluded because they operate under different geographic restrictions, making common recommendations unsuitable for both populations. Other for-hire services were also excluded as ride-hailing platforms match drivers with passengers, reducing the need for drivers to strategise independently.

1.1 Data

The primary dataset contains NYC Yellow Taxi trip records obtained from the New York City Taxi and Limousine Commission (TLC). It provides information on individual trips, including pickup and drop-off times, pickup location, fare amount and trip distance [1]. To incorporate weather conditions, weather data was obtained from the National Oceanic and Atmospheric Administration (NOAA) for four stations in the NYC area [2]. This dataset includes measures of hourly precipitation, temperature, wind speed and present weather type. The dimensions of the datasets used in the study are shown in Table 1.

Dataset	Rows	Columns
TLC Taxi Trip Records	19,785,349	19
NOAA Weather Data	70027	125

Table 1: Datasets used in the project and their shape

The period selected for this analysis spans six months, from November 2023 to April 2024. This period provides recent observations while avoiding potential confounds from the COVID-19 pandemic and the 2025 introduction of Manhattan’s congestion pricing [3]. The chosen months capture the transition from autumn to spring in NYC and are therefore, likely to provide a wide range of weather severities.

1.2 Methodology and Assumptions

Methodology: The two raw datasets are first cleaned and joined before being divided into training and test sets to prevent information leakage. A weather severity score is then constructed, followed by data aggregation and the engineering of an earnings potential measure. Geospatial and statistical analyses are then conducted on the full aggregated dataset. Finally, two regression models are developed to predict earnings potential, and their predictive performance is compared.

- Earnings potential is defined as fare revenue earned per hour spent on recorded trips (active hours) for each aggregated group. For more stable estimates, it is adjusted towards the overall earnings per active hour, with greater adjustment for groups with fewer trips and shorter trip durations.
- Weather severity is calculated using temperature, precipitation and wind speed as these weather features are most directly linked to driving conditions and demand. An equal weight is assigned to each component, with hourly dry bulb air temperature selected to represent temperature.
- Weather severity is assessed based on conditions at the trip’s pickup hour.
- Each taxi zone’s weather conditions are assumed to be reflected by observations from its nearest weather station.

2 Preprocessing

Both raw datasets may contain inconsistencies, missing values and anomalous observations. Therefore, the following steps were taken to prepare each data set for analysis:

2.1 TLC Yellow Taxi Trip Records

1. **Data Preparation:** The six monthly NYC Yellow Taxi trip files covering November 2023 to April 2024 were combined into a single dataset.
2. **Fare Structure:** Only standard rate trips (RatecodeID = 1) were retained, as non-standard rates (e.g. negotiated fares, airport rates, group rides) could introduce pricing differences unrelated to the factors being investigated [4].
3. **Pickup Zones:** Trips with pickup location IDs 264 (“Unknown”) and 265 (“Outside of NYC”) were removed, as they do not represent valid NYC taxi zones [1].
4. **Invalid Values:** Trips with non-positive distance, fare or duration were removed to exclude invalid trip records.
5. **Trip Duration:** The distribution of trip durations was examined, and the 99.9th percentile was 88.25 minutes. Therefore, trips longer than 90 minutes were considered unusually long relative to the observed data and were removed.
6. **Trip Speed:** Average trip speed was calculated from trip distance and duration, and the 99.9th percentile was 44.19 mph. Therefore, trips with average speeds above 45 mph were considered unusually high relative to the observed data and were removed.

7. **Fare Amount:** Fare amounts were inspected relative to trip distance and duration, and three trips with extremely high fares were removed.
8. **Zone Validation:** Exact duplicate trip records were removed, and pickup location IDs were verified against the NYC Taxi Zone Lookup Table.
9. **Features:** Pickup date, pickup hour, day of week and time of day variables were created from the pickup timestamp

Through this process, approximately 2,983,517 rows were removed ($\sim 15.1\%$ of the original dataset). The starting column space was reduced from 19 to 9 columns relevant to the analysis.

2.2 NOAA Weather Data

- **Data Preparation:** The yearly weather files for the four stations were combined, exact duplicate records were removed, and observations were filtered to the analysis period.
- **Feature Filtering:** Only the station, timestamp, temperature, precipitation and wind speed measurements were retained for subsequent analysis.
- **Hourly Observations:** Only hourly weather observations (FM-15) were retained so that weather conditions could be matched to each trip's pickup hour [5].
- **Weather Measurements:** Trace precipitation values (T) were converted to 0.001, and the selected weather measurements were converted to numeric format.
- **Precipitation Type:** The HourlyPresentWeatherType column was used to identify the form of the precipitation present. The detailed weather codes were grouped into broader categories of rain, snow, freezing rain, hail, thunderstorm and no precipitation.
- **Imputation:** After the train-test split, missing precipitation (2.08%) and wind speed (7.98%) values were imputed using the corresponding training set medians. Overall, only 3.35% of weather values were missing, suggesting that their impact on the results is limited.

Through this process, approximately 52583 rows were removed ($\sim 75\%$ of the original dataset). The starting column space was reduced from 125 to 9 columns relevant to the analyses.

2.3 Taxi-weather Join

Each taxi zone was mapped to its nearest NOAA weather station by calculating the distance between the zone centroid and station location. Each taxi trip was then matched to the weather observation from its assigned station using the pickup date and hour. Trips without a matching weather observation were removed. The final dataset contained 16,772,341 rows and 17 columns.

3 Analysis and Geospatial Visualisation

3.1 Distribution of Earnings Potential

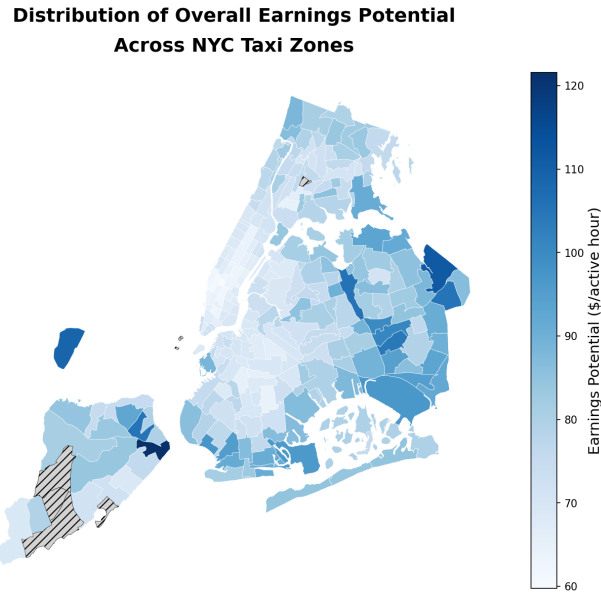


Figure 1: Spatial distribution of earnings potential across NYC taxi zones, aggregated across all weather severity levels.

Figure 1 shows that adjusted earnings potential is distributed unevenly across NYC taxi zones. Manhattan and the Bronx generally have the lowest earnings potential, with most zones appearing to fall between \$60 and \$80 per active hour. Brooklyn shows a similar pattern, although some zones along its coastal perimeters approximately fall between \$80 and \$95 per active hour. In contrast, Queens contains several higher-earning zones, some reaching close to \$112 per active hour. Staten Island generally shows lower earnings potential, with some zones having no estimate because no trip records were available. Despite this, it also contains the highest-earning zone, at approximately \$120 per active hour.

3.2 Zone-Level Impact of High-Severity Weather

Measure	Result
Zones with increased earnings	143 (57.2%)
Zones with decreased earnings	104 (41.6%)
Zones with no change	3 (1.2%)
Mean change	\$-0.65/active hour
Median change	\$+0.48/active hour
Largest increase	\$+7.87/active hour
Largest decrease	\$-27.45/active hour

Table 2: Change in earnings potential under high-severity weather relative to each zone’s overall baseline

From Table 2, we can see that the impact of high-severity weather on earnings potential varies across NYC zones. Although 57.2% of zones experience an increase in earnings potential, 41.6% experience

a decrease. This may occur because zones such as business districts maintain demand for necessary journeys during severe weather, while passengers in other zones may avoid non-essential travel. The median change is a small increase of \$0.48 per active hour, while the mean change is a decrease of \$0.65 per active hour. This indicates a greater influence of negative changes on the average. Therefore, avoiding zones that experience substantial declines may be more important than simply targeting zones with higher earnings potential.

3.3 Impact of Time of Day and Weather Severity

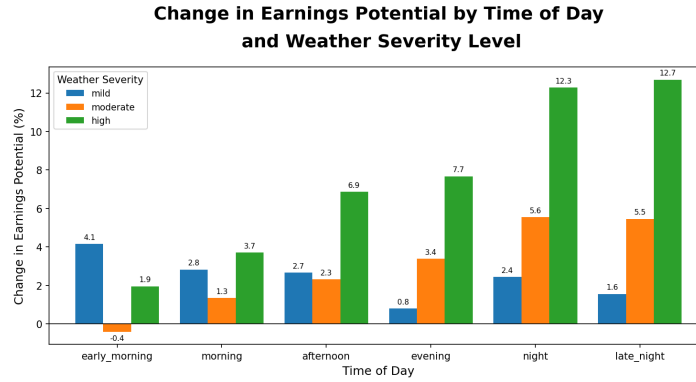


Figure 2: Change in earnings potential relative to normal severity across time-of-day and severity level.

From Figure 2, we can see that earnings potential under moderate and high-severity weather increases significantly as the day progresses. This increase is particularly pronounced under high-severity weather, rising from 1.9% in the early morning to 12.7% at late night. Therefore, the effect of weather severity on earnings potential becomes more pronounced during later hours. This pattern may be partly explained by less frequent public transport services and greater passenger safety concerns, making passengers more reliant on taxis for nighttime travel.

3.4 Impact of Precipitation Type

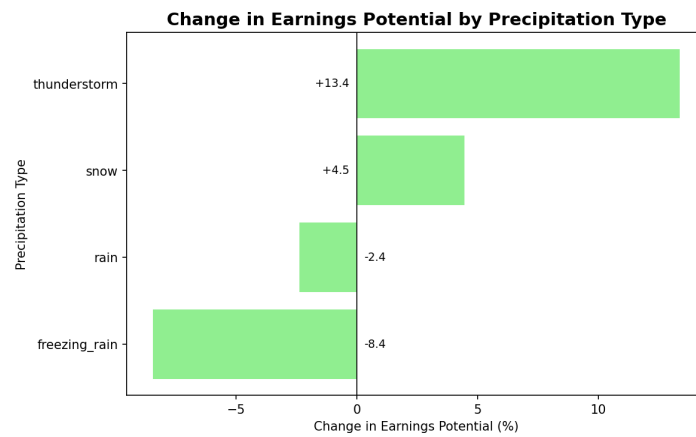


Figure 3: Change in earnings potential for each precipitation type relative to no precipitation.

From Figure 3, we can see that thunderstorms show the largest increase in earnings potential at 13.4%,

while freezing rain shows the largest decrease at 8.4%. Therefore, more severe precipitation does not always lead to higher earnings potential. While both weather conditions may lead passengers to rely more on taxis, freezing rain can cause black ice to form on roads. This may create hazardous conditions that slow trips down more than thunderstorms. Hence, drivers may spend longer travelling the same distance, reducing earnings per active hour.

4 Modelling

In this section, we build two models to predict the earnings potential (\$/active hour) for each combination of predictor values. This allows taxi drivers to compare different working opportunities as weather severity changes. The predictor variables include pickup location, time of day, week day, precipitation type and average severity score. The data is split into 80% training and 20% testing using the 80th percentile of the pickup dates (26 March 2024). Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . For improving performance and reducing the risk of overfitting, hyperparameters were tuned for both models using 3-fold cross-validation on the training data.

4.1 LASSO regression

To predict this continuous response, we first apply a LASSO regression model [6]. LASSO applies an L1 regularisation penalty that shrinks some coefficients to zero, excluding less important predictors from the model. Therefore, it allows us to determine the factors most strongly associated with earnings potential.

For this model, the categorical predictors were one-hot encoded. After fitting, the encoded categories were grouped by their original predictor, and their absolute coefficient values were summed. This provided a relative measure of importance for each original predictor.

4.2 Random Forest Regression Model

To account for potential non-linear and complex relationships between the predictors and earnings potential, a Random Forest regression model [7] was also applied. Random Forest is an ensemble method that combines multiple decision trees to make predictions. In the regression version of this model, each tree predicts a continuous numerical value rather than a class label. The final prediction is calculated as the average of the predictions from all trees.

5 Discussion

Model	RMSE	MAE	R^2
LASSO Regression	5.12	3.81	0.56
Random Forest Regression	4.02	2.93	0.73

Table 3: Test set performance of the LASSO and Random Forest regression models.

From Table 3, we can see that the Random Forest model achieved lower RMSE and MAE values than LASSO, indicating smaller prediction errors. Random Forest also has stronger explanatory power, as seen by its higher R^2 value. Hence, the results suggest that the relationship between the predictors and earnings potential is not simply linear.

Feature	Importance
PULocationID	0.9312
time_of_day	0.0363
week_day	0.0131
precip_type	0.0098
avg_severity_score	0.0096

Table 4: LASSO model Feature importance

Feature	Importance
PULocationID	0.5427
time_of_day	0.3079
week_day	0.0772
avg_severity_score	0.0505
precip_type	0.0217

Table 5: Random Forest model Feature importance

The feature importance results in Tables 4 and 5 show a similar ranking of predictors across both models. The rankings only differ in the importance of precipitation type and average severity score. In the LASSO model, precipitation type is ranked higher, whereas the Random Forest model ranks average severity score higher. The three most important predictors are pickup location, time of day and week day. This implies that where and when a taxi driver works has a stronger association with earnings potential than the weather variables.

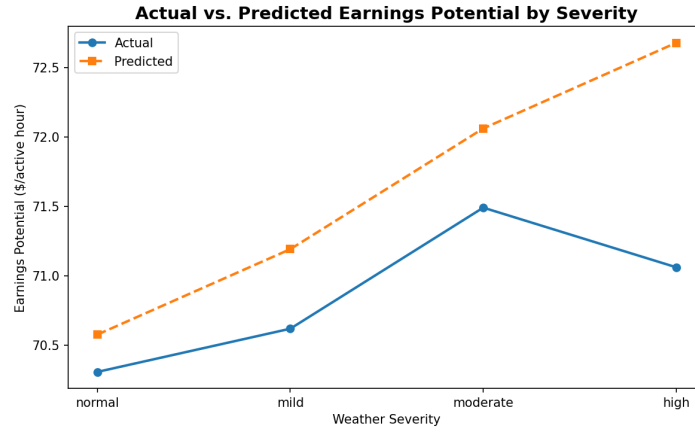


Figure 4: Average actual versus predicted earnings potential across weather severity levels.

As shown in Figure 4, the Random Forest model predictions remain relatively close to the actual earnings potential from normal to moderate weather severity. However, the model deviates the most under high-severity weather, overpredicting earnings potential by approximately \$1.59 per active hour. This indicates that the Random Forest model does not fully capture the patterns associated with the most severe conditions. This may partly be due to the smaller number of high-severity observations in the test set (4.91%). Therefore, predictions for this category may be less reliable. Further investigation

could explore additional factors that may help the model explain why actual earnings potential may not continue to increase in highly severe weather.

6 Recommendations

While severe weather may be expected to increase taxi earnings across the city, our analysis shows that this effect is not consistent across all zones. In fact, 41.6% of zones experienced decreased earnings potential during high-severity weather. Therefore, we recommend that taxi drivers prioritise relocating to zones that have historically experienced a smaller decline or an increase in earnings potential within their normal working radius. The gains from higher-performing zones are relatively small. Therefore, avoiding declining zones may be more advantageous than travelling far for higher earnings during high-severity weather. This recommendation is supported by both regression models, which identified pickup location as the most important predictor of earnings potential. We propose implementing this recommendation by providing drivers with historical data on how much earnings potential decreases in affected zones during high-severity weather. The information could be provided on the TLC website and organised by borough, allowing drivers to easily identify zones to avoid within their usual working areas.

Furthermore, our analysis suggests that the effect of increasing weather severity on earnings potential varies across time of day. Earnings potential increases most during the night and late-night hours under moderate and high severity, by up to 5.6% and 12.7% respectively (Figure 2). Therefore, we recommend that drivers with the flexibility to do so consider extending their shifts into nighttime hours when increased weather severity is forecasted. This does not require drivers to change their existing working schedule or reduce their availability during other periods. Instead, it provides an optional opportunity for drivers to earn additional income. The recommendation is also supported by both regression models, which identified time of day as the second most important predictor of earnings potential. The implementation of this recommendation only requires drivers to check publicly available weather forecasts before their shift and decide whether to extend their working hours into the night.

7 Conclusion

In this report, we investigated how weather severity affects NYC Yellow Taxi drivers' earnings potential across zones, times of day and precipitation types. The analysis found that the effect of increasing weather severity varies substantially by location and time. These findings support two practical recommendations for drivers. Firstly, drivers should prioritise avoiding zones that historically experience larger declines in earnings potential during high-severity weather. Secondly, drivers who wish to earn additional income may consider extending their shifts into nighttime hours when increased weather severity is forecasted. Together, these recommendations provide drivers with a more informed approach to adjusting where and when they work under adverse weather conditions.

References

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